

CO3-AT2-LABORATORY PERFORMANCE

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1

CODE:

Student

STUDENT

Write a C++ program to create a class for a student with a constructor and a destructor.

Constructor and a destructor.

```
1 #include <iostream>
2 #include <string>
3 using namespace std;
4
5 class Student {
6     string name;
7     int rollNo;
8 public:
9     // Constructor
10    Student(string n, int r) {
11        name = n;
12        rollNo = r;
13        cout << "Constructor called: Student object created." << endl;
14    }
15
16    // Member function to display details
17    void display() {
18        cout << "Student Name: " << name << endl;
19        cout << "Roll Number: " << rollNo << endl;
20    }
21
22    // Destructor
23    ~Student() {
24        cout << "Destructor called: Student object destroyed." << endl;
25    }
26 };
27
28 int main() {
29     Student s1("Vishnu Vardhan", 192425212);
30     s1.display();
31
32     return 0;
33 }
```

Your INPUT goes here!

OUTPUT:

OUTPUT

```
Constructor called: Student object created.  
Student Name: Vishnu Vardhan  
Roll Number: 192425212  
Destructor called: Student object destroyed.
```

2

CODE:

Book

BOOK

Write a C++ program to create a class for a book with a constructor and a destructor.

BOOK

Write a C++ program to create a class for a book with a constructor and a destructor.

```
1 #include <iostream>  
2 #include <string>  
3 using namespace std;  
4  
5 class Book {  
6 private:  
7     string title;  
8     string author;  
9     float price;  
10  
11 public:  
12     // Constructor  
13     Book(string t, string a, float p) {  
14         title = t;  
15         author = a;  
16         price = p;  
17         cout << "Book created: \"<title>\" << title << \"\"<author>\" << author << endl;  
18     }  
19  
20     void display() {  
21         cout << "Title: " << title << endl;  
22         cout << "Author: " << author << endl;  
23         cout << "Price: " << price << endl;  
24     }  
25  
26     // Destructor  
27     ~Book() {  
28         cout << "Book destroyed: \"<title>\" << title << \"\"<author>\" << author << endl;  
29     }  
30 };  
31  
32 int main() {  
33     Book b1("The Alchemist", "Paulo Coelho", 299.50);  
34     b1.display();  
35  
36     return 0;  
37 }
```

OUTPUT:

```
Book created: "The Alchemist"
Title: The Alchemist
Author: Paulo Coelho
Price: 299.5
Book destroyed: "The Alchemist"
```

3

CODE;

Rectangle

RECTANGLE

Write a C++ program to create a class for a rectangle with a constructor and a destructor.

PROGRAM EDITOR

C++

Run Save

```
1 #include <iostream>
2 using namespace std;
3
4 class Rectangle {
5 private:
6     double length;
7     double width;
8
9 public:
10    // Constructor
11    Rectangle(double l, double w) {
12        length = l;
13        width = w;
14        cout << "Constructor called: Rectangle created with length = "
15             << length << " and width = " << width << endl;
16    }
17
18    // Function to calculate area
19    double area() {
20        return length * width;
21    }
22 }
```

rectangle

RECTANGLE

Write a C++ program to create a class for a rectangle with a constructor and a destructor.

```
22
23 // Function to calculate perimeter
24 double perimeter() {
25     return 2 * (length + width);
26 }
27
28 // Destructor
29 ~Rectangle() {
30     cout << "Destructor called: Rectangle object destroyed." << endl;
31 }
32 };
33
34 int main() {
35     double l, w;
36
37     cout << "Enter length of rectangle: ";
38     cin >> l;
39     cout << "Enter width of rectangle: ";
40     cin >> w;
41
42     Rectangle rect(l, w);
43
44     cout << "Area of rectangle: " << rect.area() << endl;
45     cout << "Perimeter of rectangle: " << rect.perimeter() << endl;
46
47     return 0;
48 }
```

OUTPUT:

```
Enter length of rectangle: Enter width of rectangle: Constructor called:
Rectangle created with length = 8.8844e-312 and width = 6.95123e-310
Area of rectangle: 0
Perimeter of rectangle: 1.40802e-309
Destructor called: Rectangle object destroyed.
```

4

CODE:

Car

CAR

Write a C++ program to create a class for a car with a constructor and a destructor.

Write a C++ program to create a class for a car with a constructor and a destructor.

```
1 #include <iostream>
2 using namespace std;
3
4 class Car {
5     string brand;
6     int year;
7
8 public:
9     // Constructor
10    Car(string b, int y) {
11        brand = b;
12        year = y;
13        cout << "Car created: " << brand << " (" << year << ")" << endl;
14    }
15
16    void display() {
17        cout << "Brand: " << brand << ", Year: " << year << endl;
18    }
19
20    // Destructor
21    ~Car() {
22        cout << "Car destroyed: " << brand << " (" << year << ")" << endl;
23    }
24 };
25
26 int main() {
27     Car car1("Toyota", 2023);
28     car1.display();
29
30     Car car2("Honda", 2022);
31     car2.display();
32
33     return 0;
34 }
```

OUTPUT:

```
Car created: Toyota (2023)
Brand: Toyota, Year: 2023
Car created: Honda (2022)
Brand: Honda, Year: 2022
Car destroyed: Honda (2022)
Car destroyed: Toyota (2023)
```

5.

CODE:

Bank

BANK

Write a C++ program to create a class for a bank account with a constructor and a destructor.

constructor and a destructor.

```
1 #include <iostream>
2 using namespace std;
3
4 class BankAccount {
5 private:
6     int accountNumber;
7     double balance;
8
9 public:
10    // Constructor
11    BankAccount(int accNum, double bal) {
12        accountNumber = accNum;
13        balance = bal;
14        cout << "Constructor called for Account No: " << accountNumber << endl;
15    }
16
17    // Destructor
18    ~BankAccount() {
19        cout << "Destructor called for Account No: " << accountNumber << endl;
20    }
21
22    void display() {
23        cout << "Account Number: " << accountNumber << endl;
24        cout << "Balance: " << balance << endl;
25    }
26 };
27
28 int main() {
29     BankAccount acc1(192425212, 5000.75);
30
31     acc1.display();
32
33     return 0; // Destructor will be called automatically
34 }
```

OUTPUT:

```
Constructor called for Account No: 192425212
Account Number: 192425212
Balance: 5000.75
Destructor called for Account No: 192425212
```